



AI-Driven Classification of Virus Infections via Cytopathic Effect Imaging: A Pox Study

Yasser Taha¹, Nathalie Kinscher², Andreas Nitsche², Nils Körber¹, Claudia Kohl²

¹ Robert Koch Institute, Centre for Artificial Intelligence in Public Health Research, ZKI-PH 3, Nordufer 20, 13353 Berlin | ² Robert Koch Institute, Division of Highly Pathogenic Viruses, Seestrasse 10, 13353 Berlin

Background

Virus-infected cells undergo distinct physiological and morphological changes [1], collectively known as cytopathic effects (CPEs), which can serve as valuable indicators for virus-specific diagnostics. However, current identification methods, such as plaque assays and TCID₅₀, are **labor-intensive**, require **manual annotation**, are prone to **human error**, and can **lack virus specificity**—necessitating additional tests for accurate virus

identification. Furthermore, these techniques have **slow readout** times and rely on **chemical markers** like Crystal Violet (CV) and Hoechst. To address these limitations, an AI-based method is essential to automate CPE detection, enhance diagnostic accuracy, eliminate the need for chemical markers, and significantly speed up the analysis process.

Methodology

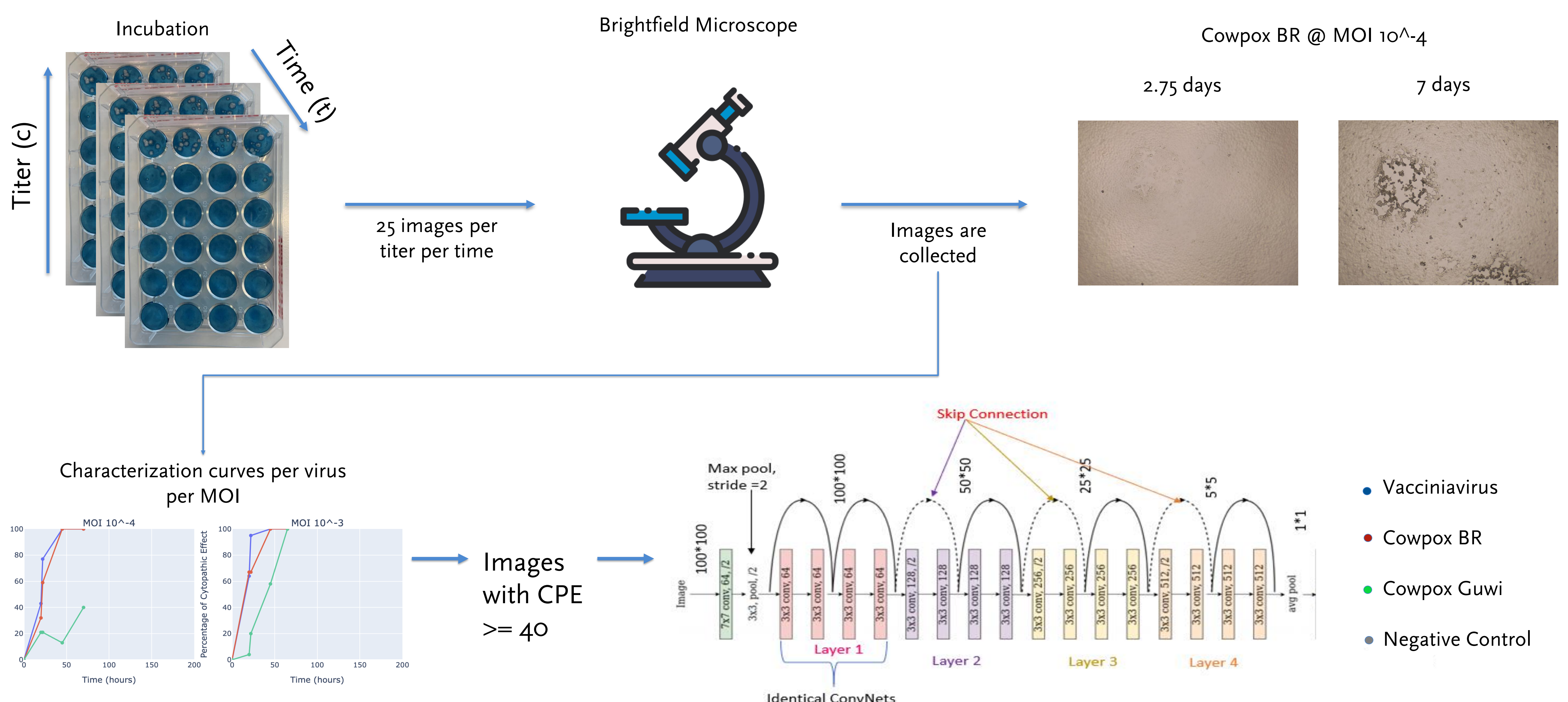


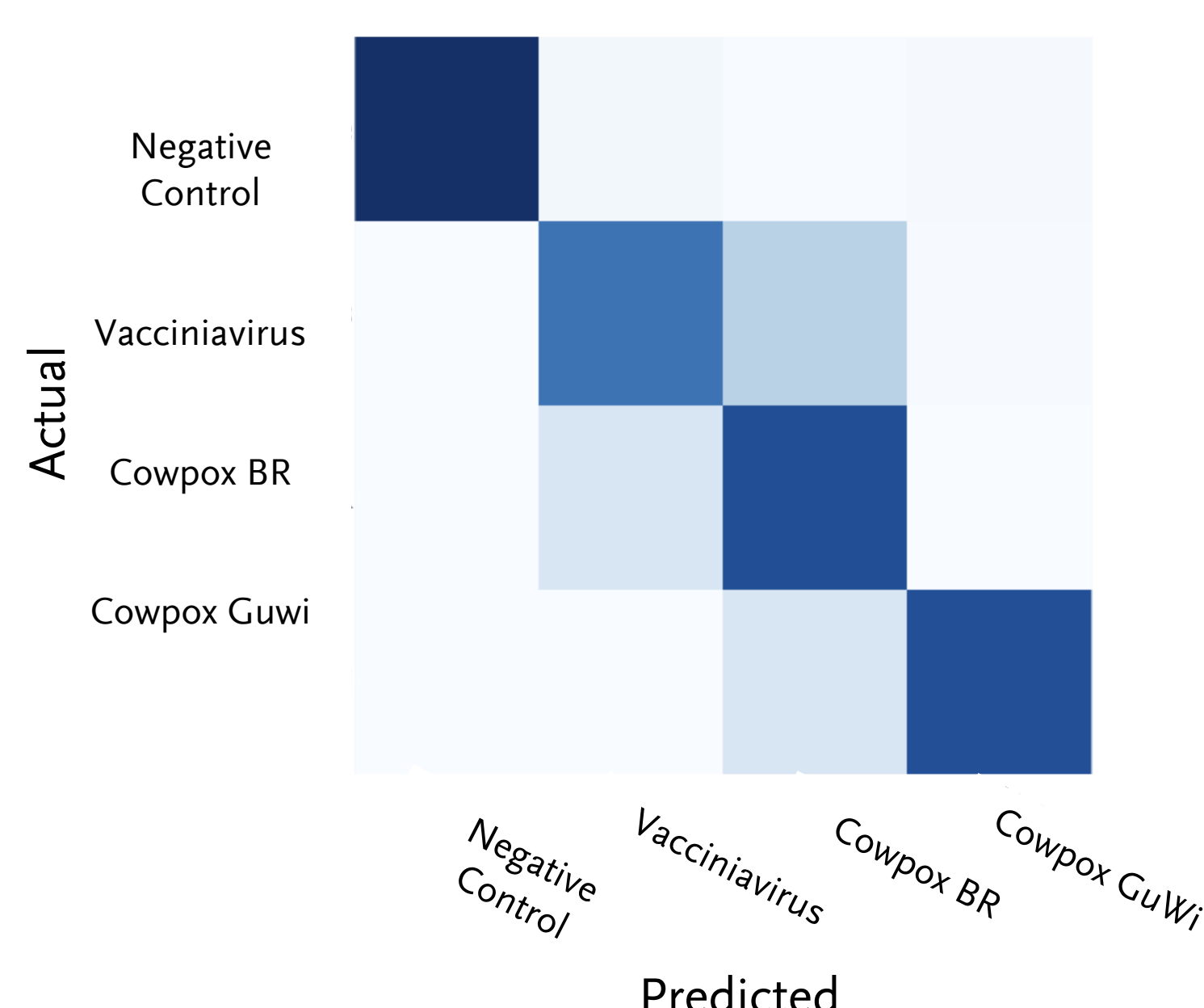
Figure 1. Cells are infected with viruses at varying concentrations and imaged over time using brightfield microscopy. Characteristic CPE curves are generated from image samples, and images from wells exceeding 40% CPE are selected for virus classification.

In this study, cell cultures are inoculated with three closely related viruses at six different multiplicities of infection (MOIs), along with a negative control. Images are captured at various time intervals, with 25 images taken per well to ensure comprehensive coverage. From this dataset, an expert annotator samples images to create characteristic CPE (cytopathic effect) curves for each virus and MOI.

These curves represent the percentage of images per well displaying CPE over time, typically increasing as the infection progresses. A threshold of 40% CPE is set to identify and select images that surpass this level, which are then used to train and evaluate a ResNet18 [2] model for virus classification.

Results and Outlook

Balanced Accuracy 83.1%



The proposed AI model achieved a balanced accuracy of 83.1% in distinguishing between closely related viruses, even at early time points where only 40% of the images in a well exhibit visible cytopathic effects (CPEs). This is particularly significant, as such early-stage CPE levels are often too subtle for human experts to reliably differentiate between viruses, demonstrating the model's potential in facilitating earlier and more accurate diagnostics.

Outlook:

- Improve model performance through failure case analysis and acquisition of additional data, including poxvirus samples
- Anomaly Detection
- Explainability
- Focus on viruses not showing human visible CPEs. Example: Hanta, Ebola, etc

Contact

Robert Koch-Institute Centre for Artificial Intelligence in Public Health Research (RKI ZKI-PH): zki-ph@rki.de
Nils Körber: koerber@rki.de
Yasser Taha: tahay@rki.de

References

- [1] Yakimovich, A. et al. Inhibition of poxvirus gene expression and genome replication by bisbenzimidazole derivatives. J. Virol. 91, e00838-17 (2017)
- [2] Kaiming He, Xiangyu Zhang, Shaoqing Ren, and Jian Sun. Deep residual learning for image recognition. In Proceedings of the IEEE conference on computer vision and pattern recognition, pages 770–778, 2016